Preliminary Palaeomagnetic Study of Igneous Rocks from Abrolhos Islands, Brazil

THERE are few palaeomagnetic data on Cainozoic rocks from the South American block[1–4], and these data refer to the Upper Cainozoic. The geological information and radiometric data indicate a Middle–Lower Cainozoic age for the igneous rocks of Abrolhos Islands[5,6]. The palaeomagnetic study of these rocks could clarify the chronology of Cainozoic geomagnetic reversals and some geological problems of the islands, and possibly allow the calculation of a Middle–Lower Cainozoic geomagnetic pole for South America.

The Abrolhos Islands are situated at 38° 42′ W and 17° 58′ S on the continental shelf of State of Bahia, Brazil. Cordani[5] and Cordani and Blasekovic[6], in their K-Ar radiometric study of the islands basaltic rocks, reviewed the previous geological work referring to the area. They suggest that the Abrolhos Islands are remnants above sea level of a volcanic episode that occurred in Middle–Lower Cainozoic times. The basaltic rocks formed during this episode overlay Cretaceous sediments dipping gently to NNW. Fig. 1 shows the schematic geological map of the Abrolhos Islands according to these authors; both coarse and fine grained basaltic rocks were observed. The radiometric study indicated K-Ar ages between 40 and 60 m.y. for these igneous rocks. The sampling places for the K-Ar study used in this palaeomagnetic study are shown in Fig. 1; they are indicated by the same symbols (AB) used by Cordani[5]. Table 1 lists these K-Ar ages; the radiometric age for the basaltic rocks from the Sueste Island is within the experimental error of that for the igneous rocks from the Siriba Island.

Sampling sites for the palaeomagnetic study (AP) are shown in Fig. 1; sixteen samples were collected by hand, at least two from each site. The measurements of the n.r.m. of these samples have been made with a spinner magnetometer (University of Buenos Aires). At least two disks from each sample have been submitted to a.c. magnetic fields in steps up to 300 oersted, to investigate the stability of its n.r.m.

The rocks from the Santa Barbara Island are coarse grained. Most of the samples collected in this island are very sensitive to a.c. cleaning and it has not been possible to isolate a stable remanent magnetization in them. Some samples are more stable but their remanent magnetization directions were inconsistent; tectonic tilting may be a possible cause for the inconsistency, as dips up to 15° were found in the Santa Barbara Island and greater dips might be possible[6]. The samples from the Guarita Island are also coarse grained and have unstable natural remanent magnetizations.

The samples collected in Sueste, Redonda and Siriba Islands are fine grained. The primary remanent magnetization of these samples was isolated by careful a.c. cleaning. Fig. 2 shows typical curves of changes in intensity of n.r.m. for specimens from these islands in response to a.c. cleaning. The consistency among the cleaned magnetization directions of samples from Sueste, Redonda and Siriba Islands can be seen in Fig. 3. Table 1 summarizes the palaeomagnetic and radiometric data.

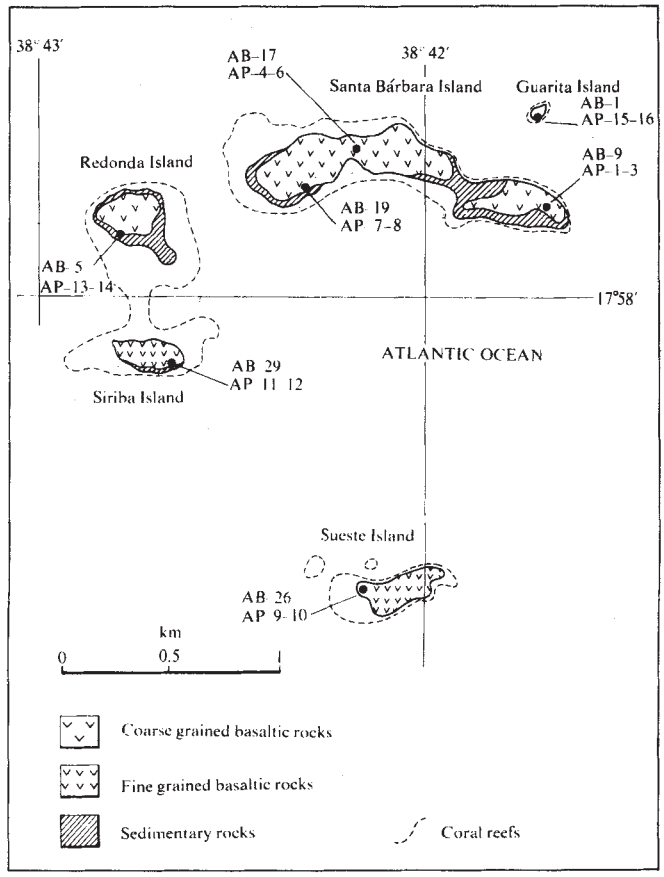


Fig. 1 Schematic geological map for Abrolhos Islands, after Cordani[5] and Cordani and Blasekovic[6]. Sample localities for the palaeomagnetic study are indicated by AP and those for the K-Ar dating by AB.

Mean directions of cleaned remanent magnetization were calculated by Fisher's statistics[7]. Virtual geomagnetic poles for each of these islands are also given in Table 1. From the mean direction of cleaned remanent magnetization of all the samples we also calculated a preliminary pole for the Abrolhos Islands (Table 1). Of course this would not be a geomagnetic pole since only three time units have been averaged and the non-dipole components of the geomagnetic field have not been completely eliminated. A further more detailed palaeomagnetic study of the Abrolhos Islands is still of interest.

The K-Ar ages of basaltic rocks from Sueste and Siriba Islands coincide (within experimental error). After cleaning, igneous rocks from the Sueste Island have remanent magnetization of reversed polarity, whereas those from the Siriba Islands have remanent magnetization of normal polarity; they were therefore formed during different volcanic events; at least, a couple of thousand years must have elapsed between these volcanic events. Unfortunately it is not possible to say which of these events was the first, but the palaeomagnetic data allow

Table 1 Summary of Palaeomagnetic and Radiometric[5,6] Data of Igneous Rocks from Abrolhos Islands, Brazil

Island	Sample numbers	K-Ar age (m.y.) (ref. 6)	Number of samples	Cleaned remanent magnetization D (°)	I (°)	α_{95} (°)	Polarity	Virtual geomagnetic poles Long. (°) (East)	Lat. (°) (South)	dψ (°)	dχ (°)
Sueste	AP 9–10	46.6±3.7	2	216.8	33	44.9	R	238	55	29	51
Siriba	AP 11–12	47.6±1.5	2	30.9	−28.5	10	N	231	60	6	11
Redonda	AP 13–14	52.4±1.6	2	210.4	31	16	R	234	61	10	18
Guarita	AP 15–16	64.5±5.8	2	—	—	—	—	—	—	—	—
Santa Barbara	AP 1–3	32.2±1.9									
	AP 4–6	37.3±2.2	8	—	—	—	—	—	—	—	—
	AP 7–8	60.7±9.1									
Abrolhos Is.	AP 9–14	—	6	212.7	30.9	6.4	N–R	234.3	59	4	7

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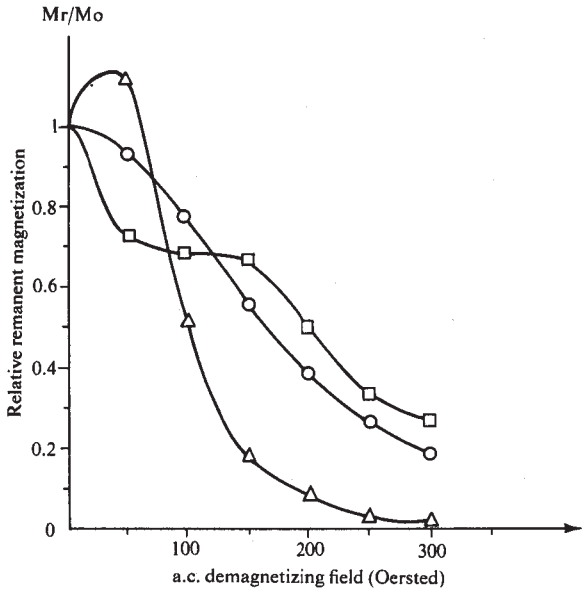


Fig. 2 Changes in intensity of n.r.m. of samples from: □, Sueste Island; ○, Siriba Island; and △, Redonda Island, in response to a.c. cleaning.

improvement of the knowledge about the volcanic cycle during which Abrolhos Islands were formed.

The errors in K-Ar ages for the rocks from Sueste, Siriba and Redonda Island used in this study are much larger than many polarity intervals of the geomagnetic field occurring within the past 4.5 m.y.[8]. Hypothetic reversal time scales going back several tens of million years have been obtained from magnetic profiles crossing active oceanic ridges, on the assumption that the rate of sea-floor spreading was constant[9,10]. Cox[8] has suggested that some of the shortest events of the geomagnetic field (shorter than 50,000 yr) were not resolved on these magnetic profiles. Then it is not possible to use the palaeomagnetic data for the igneous rocks from Abrolhos Islands to throw light on the validity of those hypothetic reversal time scales because the errors in the radiometric ages are too large.

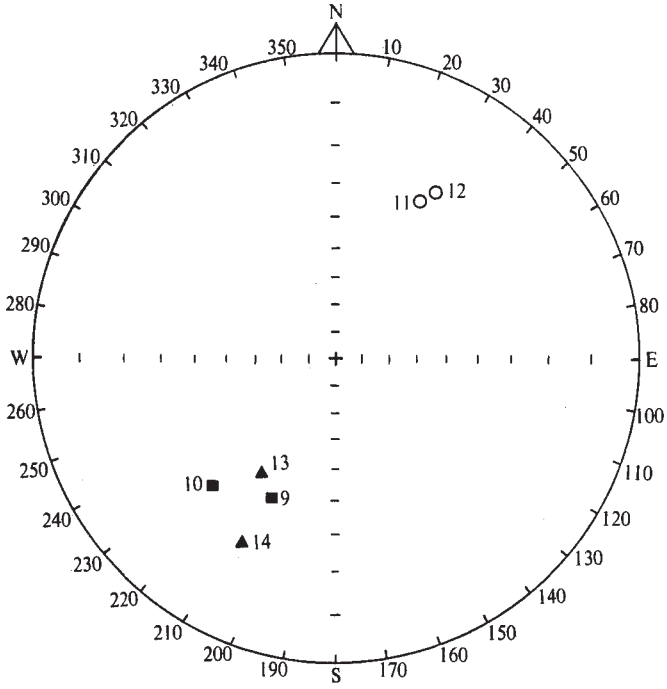


Fig. 3 Direction of cleaned remanent magnetization of hand samples from: □, Sueste Island; ○, Siriba Island; and △, Redonda Island. Downward inclinations are plotted as full symbols and upward inclinations as open circles.

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Ocean Floor Basalt Affinity of Precambrian Glaucophane Schist from Anglesey

THE Mona Complex of Anglesey and Lleyn comprises a thick geosynclinal sequence of arenaceous and argillaceous sedimentary rocks[1], which were folded, variably metamorphosed and subjected to granite intrusion during the late Precambrian[2]. The complex includes separated outcrops of spilitic lavas, basic glaucophane schists and gabbros, and serpentinized ultramafic rocks[3]. Where these rocks coexist in an orogenic environment, they have been termed an ophiolitic association. Such associations may represent thrust slices of ocean crust and upper mantle emplaced at destructive plate margins[4]. An origin for the Mona Complex at such a plate margin during the late Precambrian has been suggested by Dewey[5]. But the relationship between the different rock units in Anglesey is obscure and their affinity with oceanic crust is unproven. Here I discuss some chemical features of the glaucophane schists which suggest that these rocks have originated by metamorphism of basaltic rocks with an affinity to ocean-floor tholeiite.

The glaucophane schists outcrop over an area of 5 km² in south Anglesey and, as part of a study of the geochemistry of Monian rocks, samples were collected from the outcrops around Felinegan (G.R. SH 536735). The mineral assemblage of these rocks is: crossite+epidote±(minor) quartz, plagioclase, muscovite, chlorite, iron oxides and calcite. Lawsonite has also been reported[1] and the schists therefore belong to the glaucophane schist facies. Seven samples were analysed for major and trace elements by X-ray fluorescent methods[6] and features of these analyses are discussed in this note.

Recent studies[7,8] have attempted to show how certain trace elements may be used to determine the parental material of ophiolitic rocks. These methods may be applicable to basaltic rocks metamorphosed to lower amphibolite facies[7] and may therefore be applicable to the glaucophane schists.

Possible parental material for the basaltic components of ophiolite complexes could include: ocean-floor basalts, calc-alkaline basalt, island arc tholeiite, ocean island alkali basalt or oceanic island tholeiite.

The average major element analysis for the glaucophane schists is given in Table 1 and is compared with some average analyses of these possible parental basalt types in Table 2. Although the schist analysis resembles the subalkaline basaltic types (analyses 2–5) more closely than the alkaline basalt

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